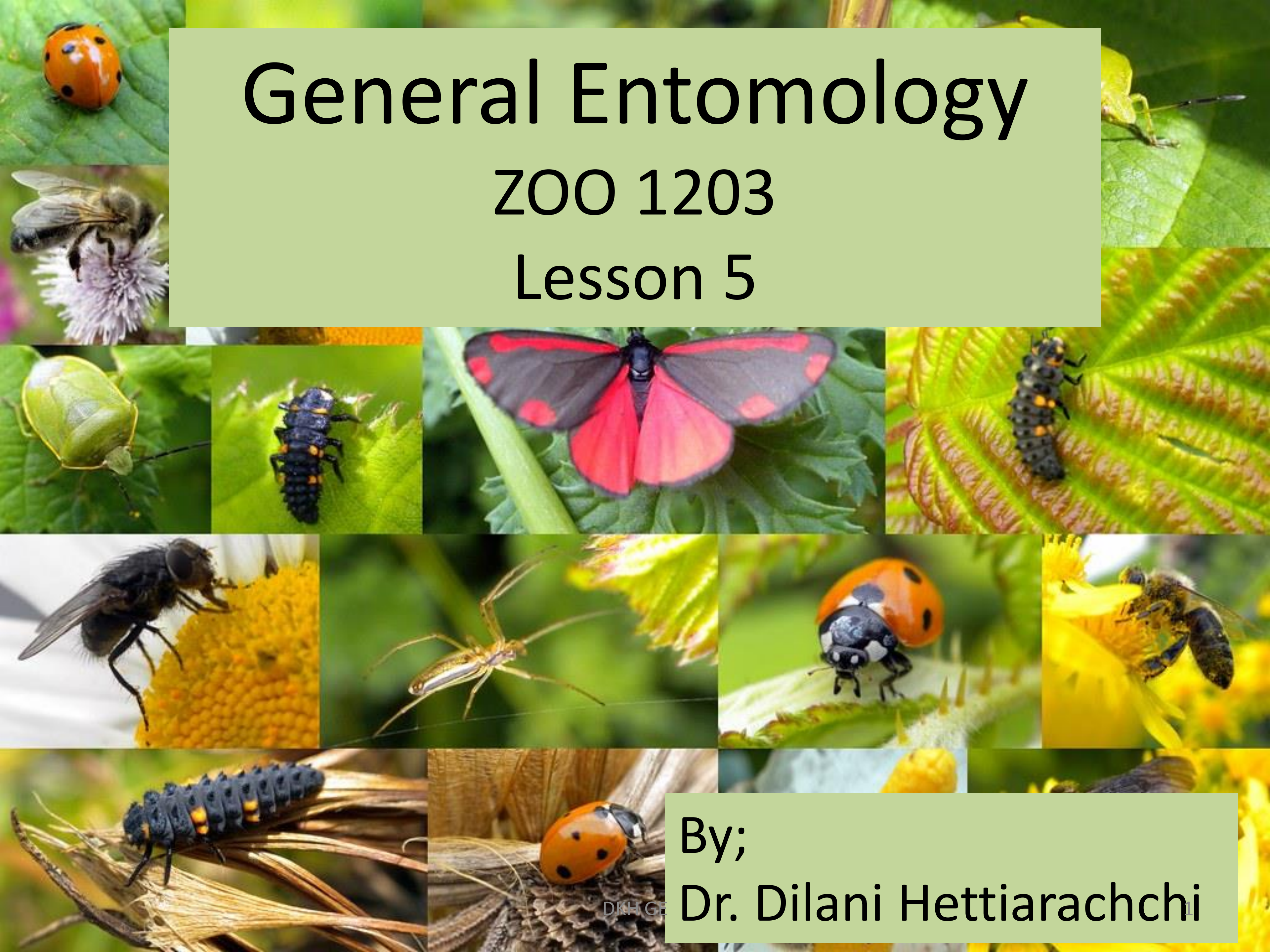

General Entomology

ZOO 1203

Lesson 5

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Lesson Objectives

After completion you should be able to;

- Describe the following systems;
    - Nervous system
    - Reproductive system
    - Endocrine system

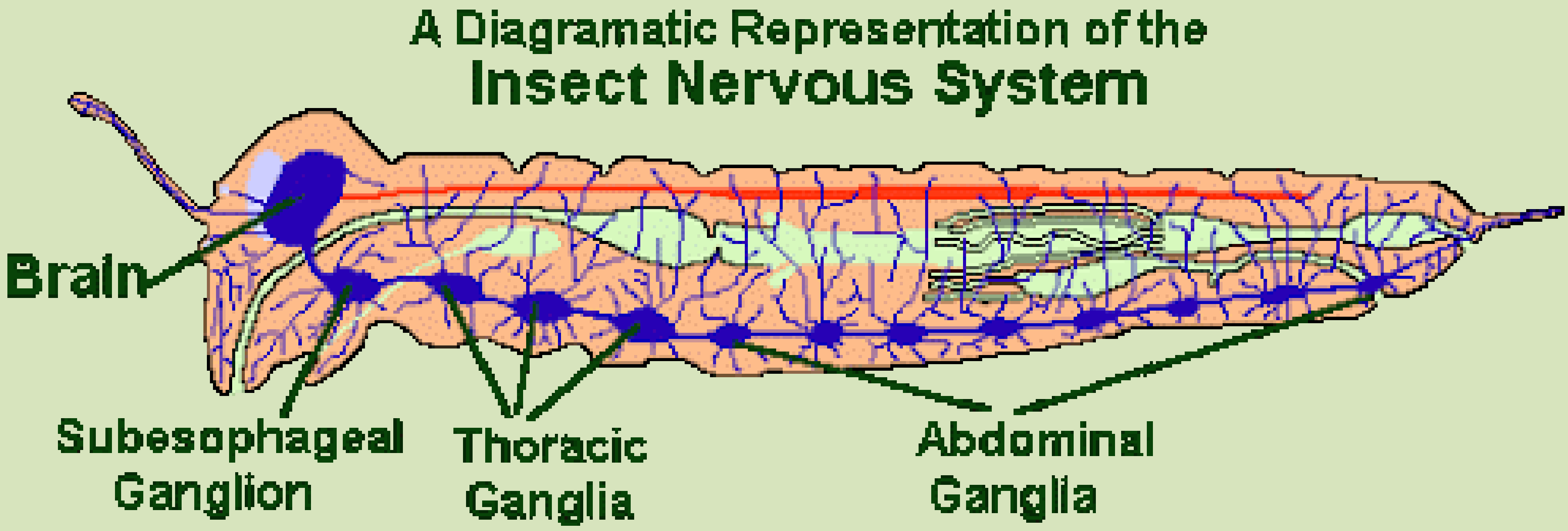
Nervous system

- Function to generate and transport electrical impulses to integrate information received and to stimulate movement

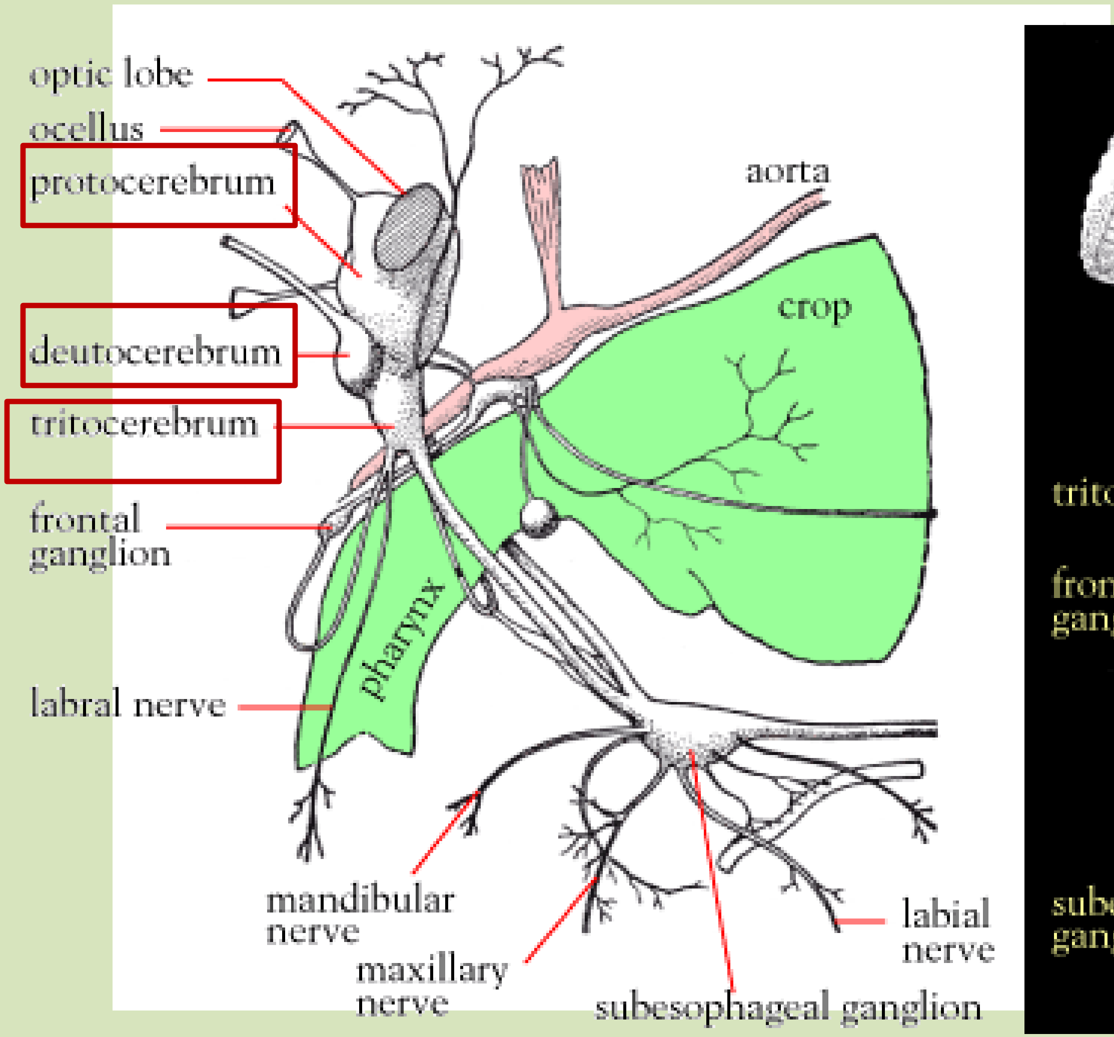
- Can be divided into two; central nervous system and Peripheral/visceral nervous system

Central nervous system

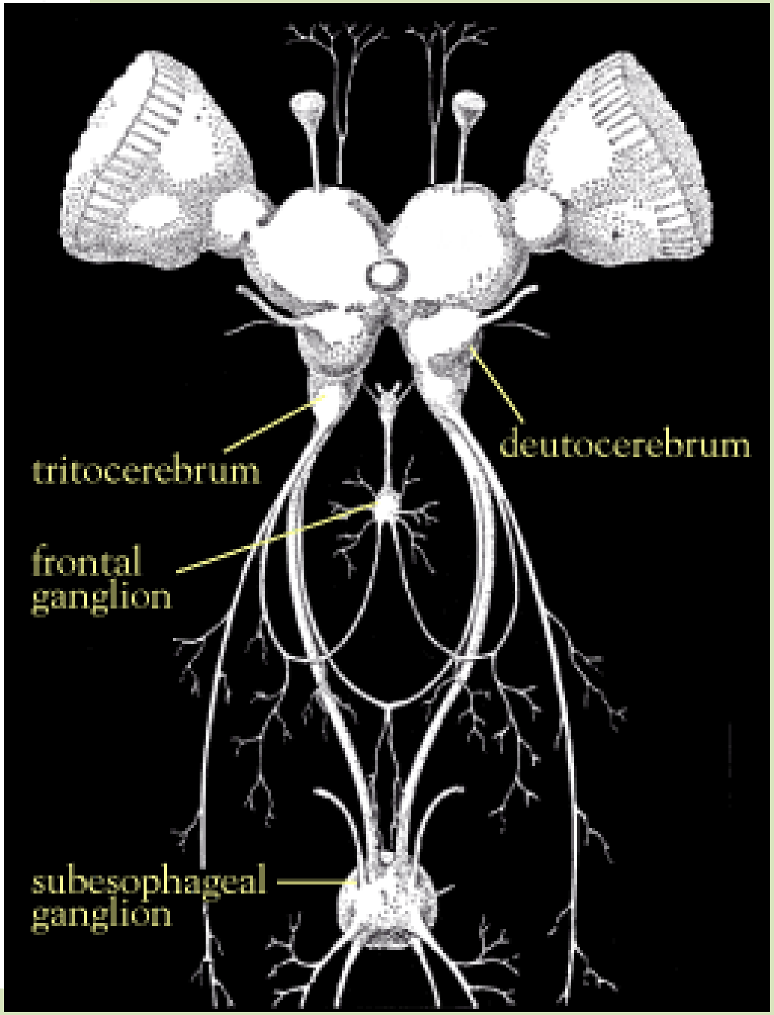
- Formed from a brain located in the head
- Ventral nerve chord runs through the body from the head
- Brain is located above the esophagous thus also called **supraesophageal ganglion**



- Brain is consisting of pair of;
  - a. protocerebrum – communicate/ innervates with compound eyes and ocelli
  - b. deutocerebrum – below the protocerebrum, innervates with the antennae
  - c. tritocerebrum – connects to the visceral nervous system



lateral view



anterior view

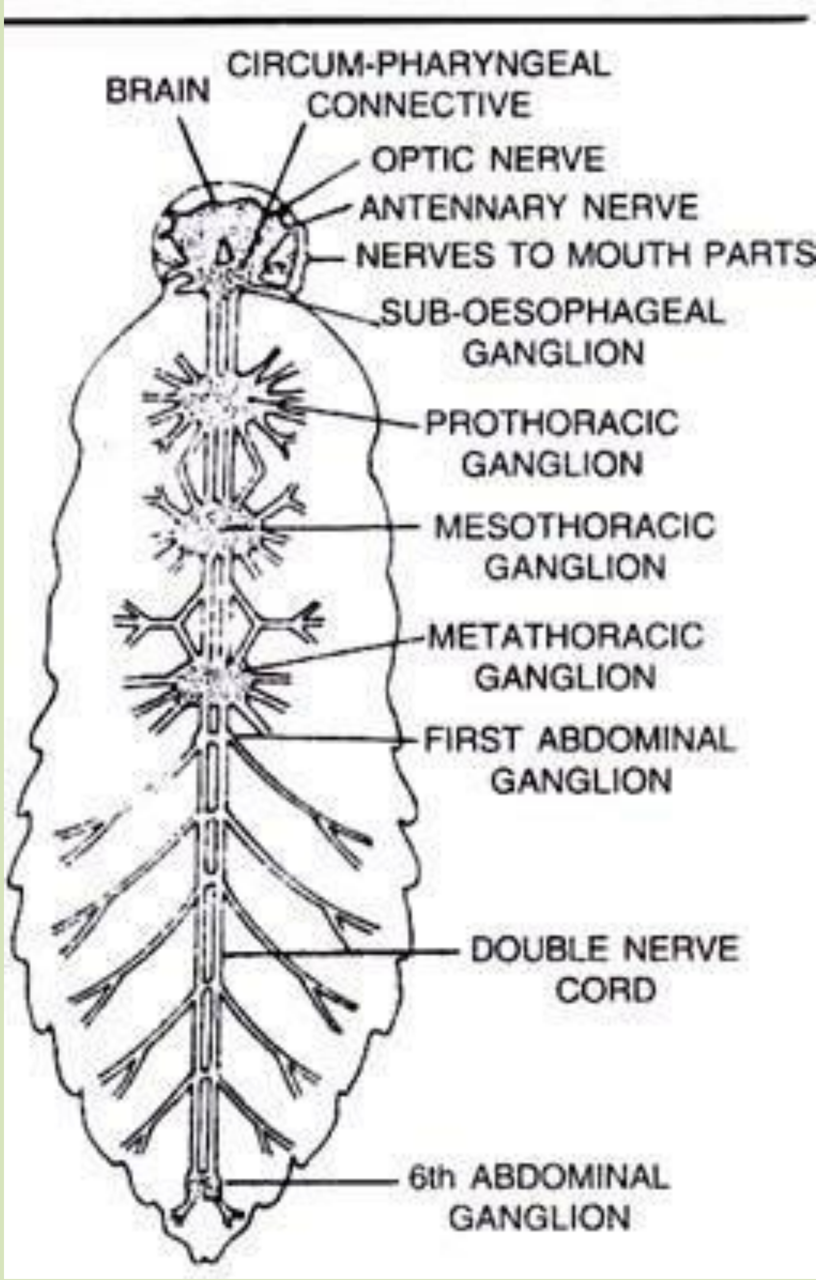
Central nervous system of a grasshopper

- Brain connects to a large nerve ganglion located under the esophagus – **subesophageal ganglion**

- Subesophageal ganglion → innervates the mouthparts and the salivary ducts

- Ventral nerve chord composed of a series of ganglia

- Typically there is a ganglion in each thoracic segment and in many, one in each abdominal segments



- Nerves arise from each ganglion and branch and innervates both sensory structures and muscles

Visceral nervous system

- Consists of frontal ganglion connected to the brain and other small ganglia

- Ganglia give rise to paired nerves

- They innervate mainly the digestive system and endocrine glands; **corpora cardiaca** and **corpora allata**

- Both glands involved in insect growth

- It also connects with other systems

Nerves/ neurons

- Basic unit of nervous system
- Neuron is composed of a cell body, one or more receptor fibrils and an axon
- Three types of neurons;
  1. sensory neurons – connect to sense organs, carries impulse to ganglia in the CNS
  2. motor neurons – connect with muscle tissues, carries impulse away from CNS
  3. interneurons- connects sensory and motor neurons

- Impulses arise from the flow of positive sodium ions into the cells

- And it stops quickly

- Na K pump is involved

- Na is flowed inside and K outside of the cell

- This transmission is named **axonic transmission**

- Apart from this **synaptic transmission** also take place

- A synapse is a junction between neurons and other cells i.e. interneurons and a sensory neuron/ interneuron and a motor neuron

- Acetylcholine is there as a chemical transmitter and enzyme acetylcholinesterase is also present

- Brain is the coordinating center

- Subesophageal ganglion is the coordinating center for feeding

- Each thoracic and abdominal ganglion organizes and coordinates the behaviour of its own segment

Reproductive system

- Power of reproduction is their most important biological feature

- Most insects are dioecious (male and female present)

Female Reproductive system

- Main organs are **paired ovaries**
- Each ovary is made up of bundle of several tubular **ovarioles**
- Eggs are formed there
- Each ovariole is attached with a thread called the **terminal filaments**

- Germ cells form in the ovariole just below this filaments

- Germ cells develop as they move along and arrive as fully formed eggs at the ovariole base/ pedicel

- Eggs pass into the paired lateral oviducts

- Then to the common oviduct

- From the oviduct they move to the vagina

- Eggs fertilized there and held for laying

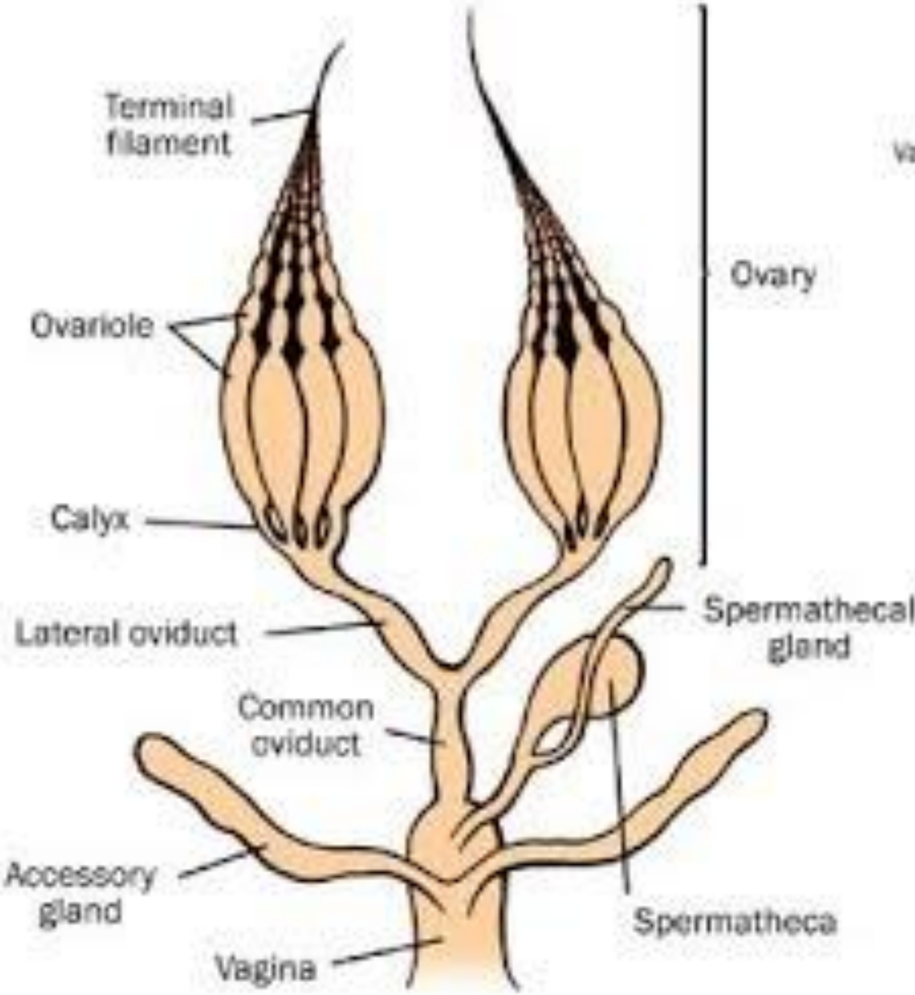
- Spermatheca receives and store sperms after copulation

- Spermathecal glands supply nutrients to maintain sperm before it is dispensed

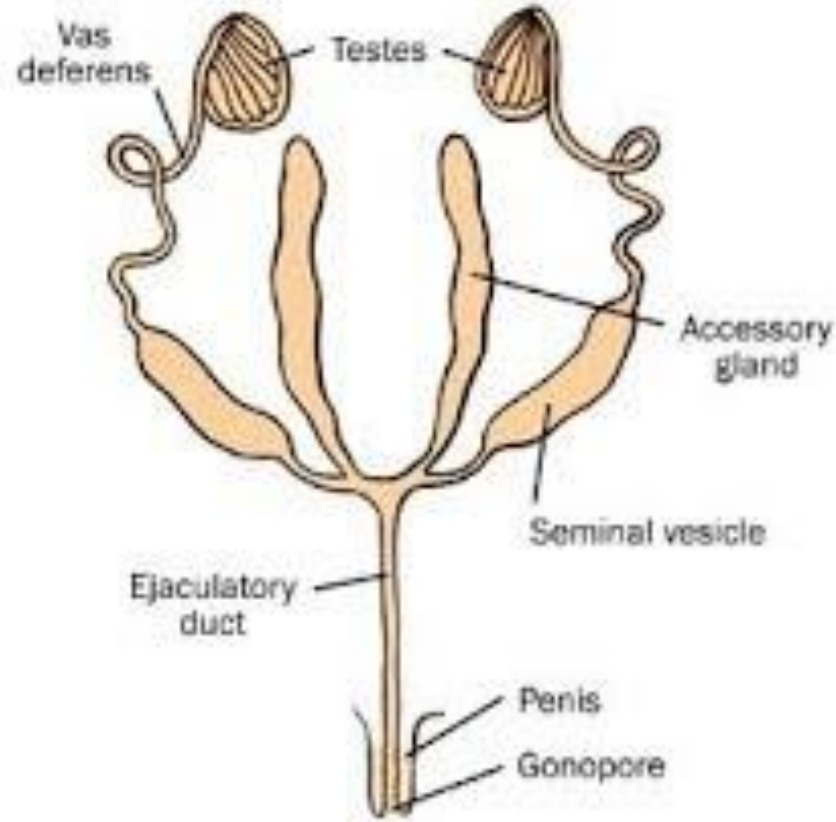
- Paired accessary glands secrete adhesive and protective covering for the egg after it has been fertilized

- Many modifications of this basic system occur among insect groups

Sperm Development



Female



Male

Male Reproductive system

- Primary organs are **paired testes**

- Each testis is made up of a number of sperm tubes enclosed in sheath

- Sperm are produced in the **sperm tubes**

- Move through the narrow **vasa efferentia**

- It empty into a common duct **vas deferens**
- Sperms move through the vas deferens and held in the seminal vesicle; the storage organ

- Paired accessary glands connect with seminal vesicles which secrete semen

- During copulation semen from the seminal vesicles moves through the ejaculatory duct and out the penis

- Aedeagus is a structure located to aid in the copulatory process – situated near the ejaculatory duct

Endocrine system

- Hormone → chemical messengers
- Typically produced in very small quantities
- But may cause profound changes in their target cells
- Effect maybe stimulatory or inhibitory
- In some cases; single hormone may have multiple targets and cause different effects in each target

- There are at least 4 categories of hormones-producing cells in an insect body

1. Endocrine glands – adapted exclusively for producing hormones and releasing them into the circular system

2. Neurohemal organs – similar to glands. But they store their secretory product in a special chamber until stimulated to release it y a signal from the nervous system/other hormone

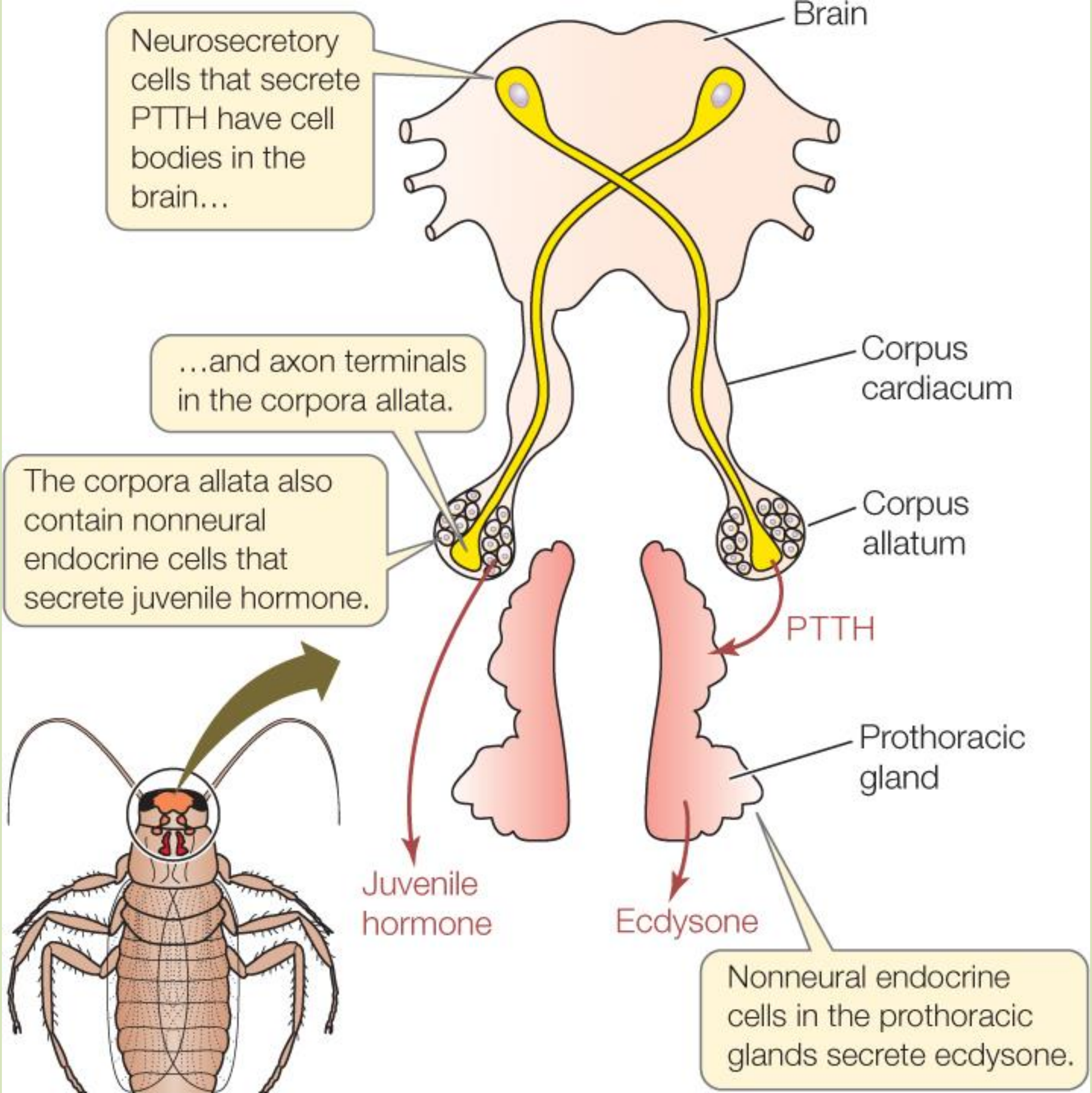
3. Neurosecretory cells – specialized nerve cells that respond to stimulation by producing and secreting specific chemical messengers. They serve as a link between the nervous and endocrine stystem

4. Internal organs – hormone-producing cells are associated with numerous organs of the body, including the ovaries and testes, the fat body and parts of the digestive system

- Together these forms the endocrine system

Prothorasic gland

- Largest gland
- Just behind head
- Manufacture ecdysteroid / molting hormones (steroid hormones including ecdysone)
- This stimulate the synthesis of chitin and protein in epidermal cells ends in molting
- Once an insect become an adult, this gland will wither away (atrophy) and insect will never molt again



Brain
Neurosecretory cells that secrete PTTH have cell bodies in the brain…
…and axon terminals in the corpora allata.
Corpus cardiacum
The corpora allata also contain nonneural endocrine cells that secrete juvenile hormone.
Corpus allatum
PTTH
Prothoracic gland
Juvenile hormone
Ecdysone
Nonneural endocrine cells in the prothoracic glands secrete ecdysone.

- Prothoracic gland is only stimulate if **Prothoracicotropic hormone (PTTH)** is present

- It is a peptide hormone secreted by **corpora cadiaca/cardiacum (singuglar)**

- Corpora cadiaca is a nurohemal organ located on the walls of the aorta just behind the brain

- Corpora cardiaca only release PTTH if it receive a signal from neurotransmitory cells in the brain

- The **corpora allata** – nurohemal organ, lie just behind the corpora cardica

- It manufacture **Juvenile hormones (JH)**

- JH is a compound that inhibits development of adult characteristics during immature stages

- It also promotes sexual maturity during the adult stage

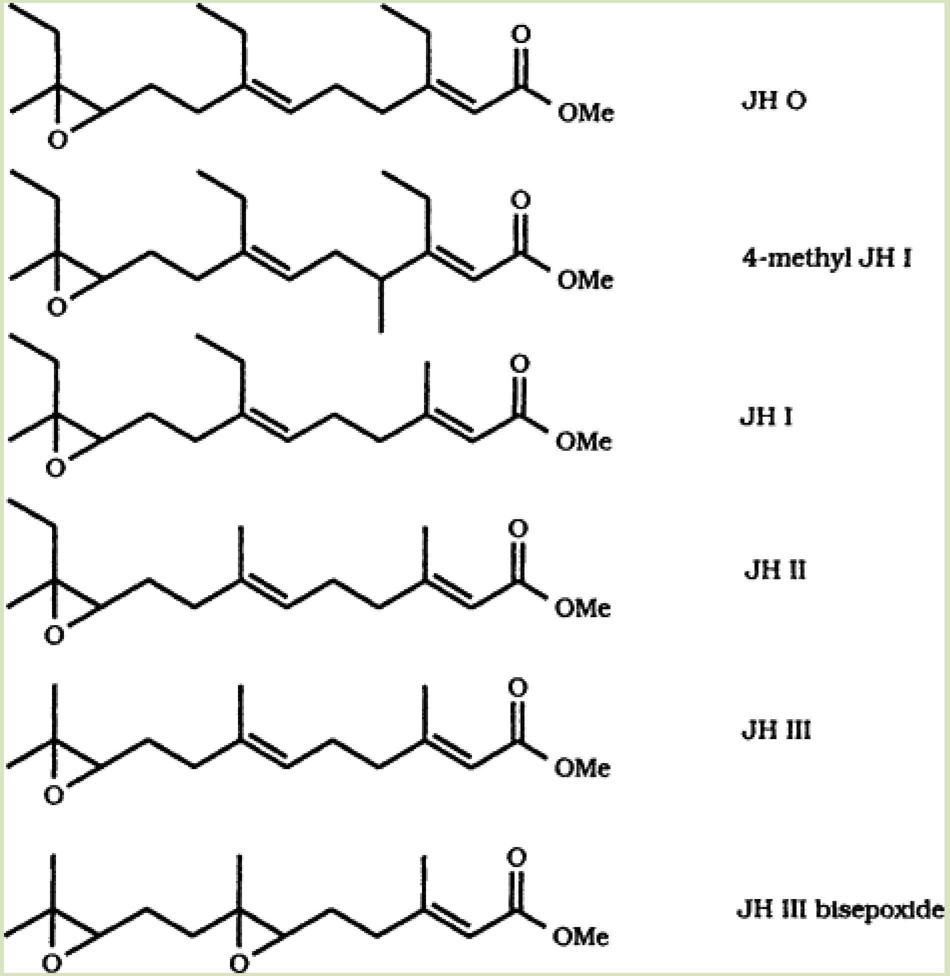
- Neurosecretory cells in the rain regulate activity of the corpora allata

- It stimulate them to produce JH during larval and nymphal stages and inhibit during transition to adulthood

- JH production is stimulated once the adult is ready for reproduction

- Chemical structure of JH is unusual (found in six different forms)

- It is a sesquiterpene compound which is more similar to defensive chemicals in pine trees than to any other animal hormone



- Neurosecretory cells - found in clusters in medial and lateral sides of the insect brain

- Connected with corpora cadiaca and corpora allata by axons

- Produce and secrete brain hormones – low molecular weight peptide that appear to be similar to PTTH

- Large number of neurosecretory cells occur in the ventral ganglia of the nerve cord

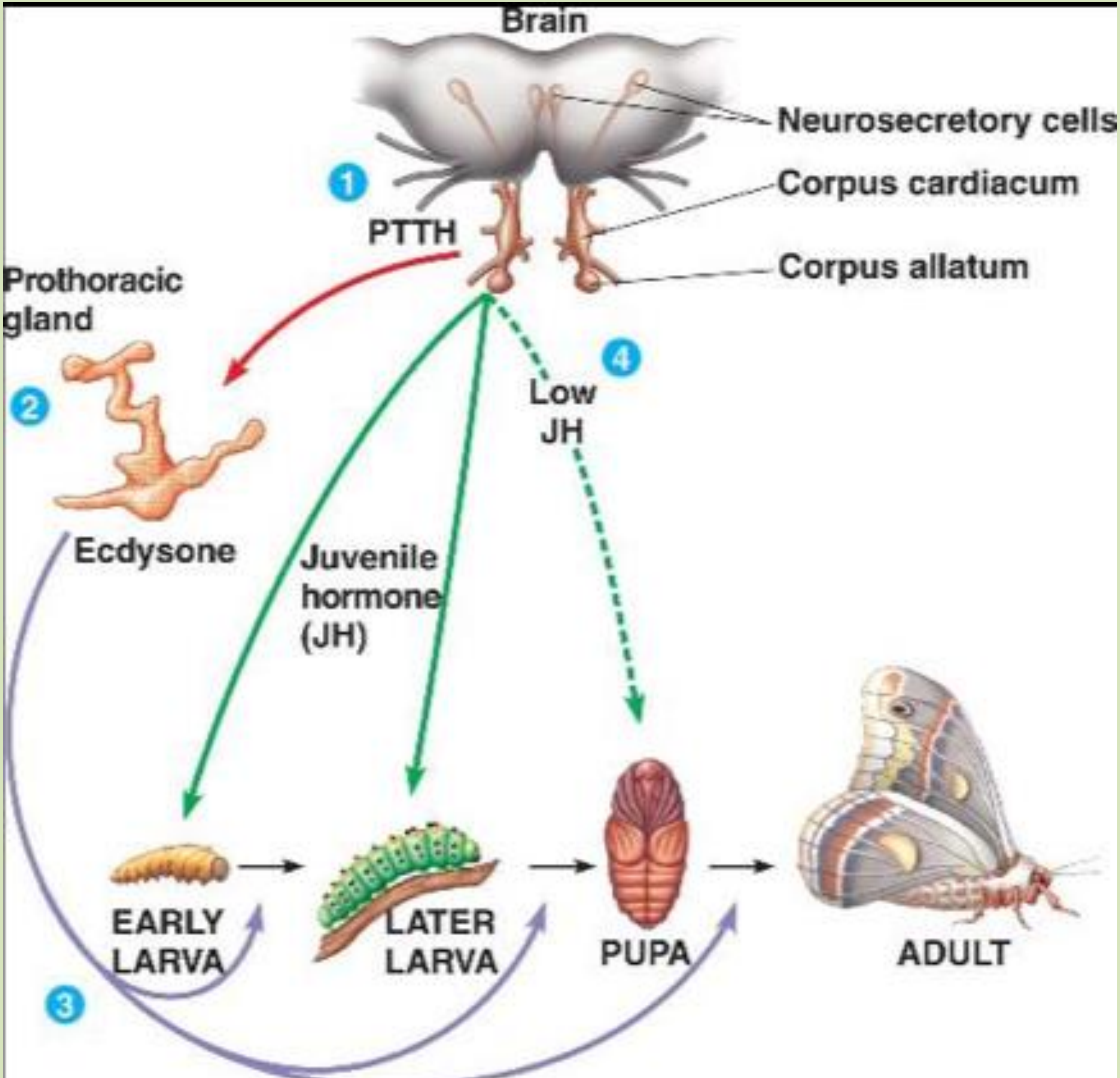
- Many other tissues and organs produce hormones

- Ovaries and testes – produce gonadal hormones → coordinate courtship and mating behaviours

- Ventral ganglia produce eclosion hormone – helps the insect to shed its old exoskeleton

- Also bursicon – hardening and tanning of the new cuticle

- There are other hormones that:
  - →control the level of sugar dissolved in blood
  - → adjust salt and water balance
  - → regulate proteins


Brain
Neurosecretory cells
Corpus cardiacum
Corpus allatum
1
PTTH
Prothoracic gland
2
4
Low JH
Ecdysone
Juvenile hormone (JH)
EARLY LARVA
LATER LARVA
PUPA
ADULT
3

Refer:

Hormonal control of molting and metamorphosis


THANK YOU!